MTH2008 – Real Analysis

2026/27

0 – Introduction

Who am I

Dr Tony Samuel – https://tony-samuel.github.io/

- Lecturer in Mathematics
  University of Exeter
- Deputy Director of Business Engagement and Innovation
  University of Exeter
- Senior Honorary Research Fellow
  University of Birmingham
- Knowledge Exchange Champion
  Knowledge Exchange Hub for the Mathematical Sciences
- Advisory board
  Academy for the Mathematical Sciences
- Peer Review College
  Engineering and Physical Sciences Research Council

Motivation

- Real Analysis is where mathematics becomes rigorous.
- Up until now, you have used calculus, limits, and infinite series because they worked.
- Real Analysis asks: *why do they work?* It gives us the solid foundations behind every formula you have learned.
- By the end, you will not only know new theorems, but hopefully, you will see mathematics itself in a sharper, more beautiful light.
- Applications of Real Analysis in my own research includes:
  - image based cancer diagnostics, and
  - ultrasound signal analysis for diagnosing and predicting battery cell defects.
- Other applications include:
  - building LLM's,
  - stochastic processes which are used in finance, AI and data science,
  - signal processing, and
  - thermodynamics, quantum mechanics, and electromagnetism.

Learning outcomes

Module Specific Skills and Knowledge

- State and prove key theorems in real analysis using a rigorous approach.
- Develop proofs related to topological concepts such as limits and connectedness.
- Understand the basis of integration of functions of a real variable.

Discipline Specific Skills and Knowledge

- Apply fundamental mathematical concepts, manipulations and results in analysis.
- Formulate rigorous arguments as part of your mathematical development.

Transferable/Employment Skills and Knowledge

- Think analytically and use logical argument and deduction.
- Communicate your ideas effectively in writing and verbally.
- Manage your time and resources effectively.

Course structure

- There are 3 lecture slots per week:
  - Monday 11:30-12:30
    Amory – Parker Moot Lecture Theatre
  - Tuesday 08:30-09:30
    Streatham Court – Lecture Theatre A
  - Thursdays 15:30-16:30, repeated 16:30-17:30
    Queens – Lecture Theatre 1
- There are 2 Graded Formative Assessments
  These do not count towards your final grade,
  but if you would like feedback on your work, these are to
  be submitted in Weeks 3 and 8 via ELE.
- There are 5 tutorials happening on the Wednesday of
  Weeks 2, 4, 7, 9 and 11
  - Tutorial 1 (Week 2): Exercise Sheet 1
  - Tutorial 2 (Week 4): Graded Formative Assessment 1
  - Tutorial 3 (Week 7): Exercise Sheet 2
  - Tutorial 4 (Week 9): Graded Formative Assessment 2
  - Tutorial 5 (Week 11): Exercise Sheet 3
- The assessment for the course will be
  - an on-campus closed book in-person written 2-hour
    final exam worth 70% of your final grade,
  - 2 Summative Assessment,
    each worth 15% of your final grade
    to take place in-class on 20 Oct. and 24 Nov.
- Office hours
  Tuesdays 09:45-11:15 (Harrsion – Room 205)
  in Weeks 2, 3, 5, 7, 8, 9, 10 and 11.

1 – The Real Numbers

We begin the study of the real number system.

- § 1.1 deals with the axioms that define the real numbers, definitions based on them, and some basic properties that follow from them.
- § 1.2 introduces ideas of set theory in the context of sets of real numbers. Here, we prove two fundamental results:
  - **Heine–Borel Theorem** and
  - **Bolzano–Weierstrass Theorem**.

1.1 The Real Number System

We denote the set of **real numbers** by the symbol $\mathbb{R}$, on which we have the binary operations of addition $(+)$ and multiplication $(\cdot)$. The real numbers $\mathbb{R}$ equipped with these operations form what is know as a **field**, specifically they satisfy the following operations:

↳ critical in elliptic curve cryptography

(a) **Commutativeity**
$a + b = b + a$ and $ab = ba$ for all $a, b \in \mathbb{R}$

(b) **Associativeity**
$(a + b) + c = a + (b + c)$ and $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all $a, b, c \in \mathbb{R}$

(c) **Distributiveity**
$a \cdot (b + c) = a \cdot b + a \cdot c$ for all $a, b, c \in \mathbb{R}$

(d) **Additive identity**
$0 + a = a$ for all $a \in \mathbb{R}$
Multiplicative identity
$a \cdot 1 = a$ for all $a \in \mathbb{R}$

(e) **Additive inverse**
For $a \in \mathbb{R}$ there is a real number $-a$ with $a + (-a) = 0$
Multiplicative inverse
For $a \in \mathbb{R}$ with $a \neq 0$, there is a real number $1/a$ so that $a \cdot (1/a) = 1$.

A set on which two binary operations are defined so as to have properties (a)–(e) is called a **field**. The real number system is by no means the only field.

Other examples of fields.

- $\mathbb{Q}$ - the rationals with the usual notions of addition and multiplication.

Recall: A rational number is a number $\frac{p}{q}$ where $p, q \in \mathbb{Z}$ - the integers and $q \neq 0$.

- $\mathbb{Z}_2$ - the set $\{0,1\}$ where we define addition by

$0+1=1 \qquad 1+1=0$

and multiplication by

$0 \cdot 1 = 0 \qquad 1 \cdot 1 = 1$

- Try proving that these two examples satisfy all of the properties to the left.

The order relation

Not all sets have an ordering. For instance think about the set comprising of an orange, an apple and a pear – there is no natural ordering on this three element set.

However, the real numbers are ordered by the relations

$$< \quad > \quad \leq \quad \geq .$$

Indeed, in addition to the field properties, we have:

(f) For each pair of real numbers a and b, exactly one of the following is true

$$a = b, \quad a < b, \quad \text{or} \quad b < a.$$

(g) If $a < b$ and $b < c$, then $a < c$ (Transitivity)

(h) If $a < b$, then $a + c < b + c$ for $c \in \mathbb{R}$
If $a < b$, then $a \cdot c < b \cdot c$ for $c \in \mathbb{R}$ with $c > 0$

A field with an order relation satisfying (f)–(h) is called an **ordered field**.

Thus, the real numbers together with the binary operations of addition and multiplication, and the order relation $<$ form an ordered field.

- Here we might write such a set as $\{\text{apple}, \text{orange}, \text{pear}\}$.

- $\mathbb{Q}$ with the usual notion of addition and multiplication and the order relation $<$ is an ordered field.
In (h) we replace $\mathbb{R}$ by $\mathbb{Q}$.

- $\mathbb{Z}_2$ with the notions of addition and multiplication defined above although a field cannot be made into an ordered field.
In (h) we replace $\mathbb{R}$ by $\mathbb{Z}_2$.

To prove this statement we first observe that if there was an order we would have two options: (i) $0 < 1$ or (ii) $1 < 0$

But if (i) occurs $0 + 1 = 1$ and $0 \cdot 1 = 0$ $\Rightarrow$ by (h) we would need to have $1 < 0$ $\Rightarrow$ $\lightning$

Try showing that a similar contradiction occurs if we assume (ii). ∎

Theorem (The triangle inequality)
If a and b are any two real numbers, then

$$|a+b| \leq |a| + |b|.$$

This is one of the most important inequalities in mathematics.

Proof: We observe that there are 4 possibilities.

(i) if $a \geq 0, b \geq 0$. then $a+b \geq 0 \Rightarrow |a+b| = a+b = |a|+|b|$.

(ii) if $a \leq 0, b \leq 0$, then $a+b \leq 0 \Rightarrow |a+b| = -(a+b) = -a+-b = |a|+|b|$.

(iii) if $a \geq 0, b \leq 0 \Rightarrow a+b = |a|-|b|$

(iv) if $a \leq 0, b \geq 0 \Rightarrow a+b = -|a|+|b|$

$$|a+b| = \begin{cases} |a|-|b| \leq |a|+|b| & \text{if } |a| \geq |b| \\ |b|-|a| \leq |a|+|b| & \text{if } |a| \leq |b| \end{cases}$$

■

Corollary (The reverse triangle inequality)
If a and b are any two real numbers, then

$$|a-b| \geq ||a|-|b|| \quad \text{and} \quad |a+b| \geq ||a|-|b||.$$

The trick of adding zero or multiplying by one is one of the most powerful and useful proof techniques.

Proof: $|a| = |a-b+b| = |(a-b)+b|$

By the triangle inequality $\leq |a-b| + |b|$

$\Rightarrow |a-b| \geq |a|-|b| \Rightarrow |a-b| = |b-a| \geq |b|-|a|$

by swapping the roles of a and b.

$\Rightarrow |a-b| \geq ||a|-|b||$ or $|a-b| \geq \max\{|a|-|b|, |b|-|a|\}$.

Replacing b by $-b$ in the above arguments yields the second inequality. ■

Supremum of a set

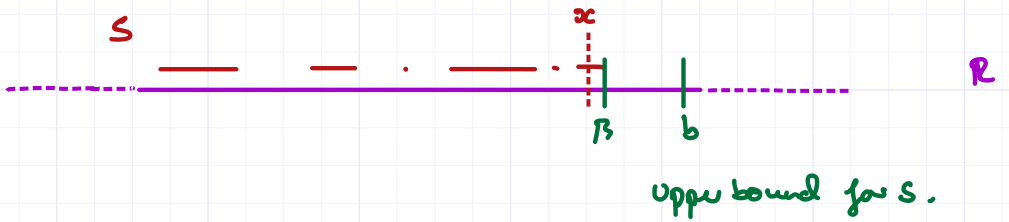
A set S of real numbers is **bounded above** if there is a real number b such that $x \leq b$ whenever $x \in S$.

In this case, b is called an **upper bound** of S.

If b is an upper bound of S, then so is any larger number because of the property of transitivity of our order relation $<$.

If β is an upper bound of S, but no number less than β is, then β is a **supremum** of S, and we write

$$\beta = \sup S.$$



upper bound for S.

i.e. β is the least upper bound of S.

Note: β does not necessarily belong to S.

Example

- If S is the set of negative numbers then any non-negative number is an upper bound for S. e.g. 42 is a non-negative number

  Since the smallest non-negative number is 0, this implies if $x \in S$ then $x < 0$

  but $\sup S = 0$. => since $0 < 42$, then $x < 0 < 42$.

- If S_1 is the set of negative integers, then any number $a \geq -1$ is an upper bound for S_1 and $\sup S_1 = -1$.

Defn: A non-empty set is a set that has at least one member.

The empty set, denoted by $\emptyset$, is the set that has no members. ← A very important concept.

The Completeness Axiom

It is one thing to define an object and another to show that it always exists. For example:

- Does it make sense to define the smallest positive real number?
- Does the empty set, as a subset of $\mathbb{R}$ have a supremum?

Since the property that a subset of $\mathbb{R}$ has a supremum, although it might not belong to the set, is indispensable in the rigorous development of calculus, we take it as an axiom for the real numbers.

(i) If a nonempty set of real numbers is bounded above, then it has a supremum.

This property is called **completeness**, and we say that $\mathbb{R}$ together with the binary operations of addition and multiplication, and the order relation $<$ is a complete ordered field.

In fact it can be shown that the real number system is essentially the only complete ordered field.

Theorem
If a nonempty set S of real numbers is bounded above, then $\sup S$ is the unique real number β such that

(1) $x \leq \beta$ for all x in S, and

(2) if $\epsilon > 0$, no matter how small, there exists an x_0 in S with $x_0 > \beta - \epsilon$.

Proof: We first show that $\beta = \sup S$ has properties (1) and (2) and then show uniqueness.

Since, by definition, β is an upper bound for S it satisfies (1).

Moreover, since any real number less than β can be written as $\beta - \varepsilon$ for some $\varepsilon > 0$ if there existed no $x_0 \in S$ with $x_0 > \beta - \varepsilon$ $\Rightarrow$ $\beta - \varepsilon$ is an upper bound for S but $\beta = \sup S$ is the least upper bound $\Rightarrow$ ↯

Thus, $\beta = \sup S$ satisfies (2).

For uniqueness, suppose there exists $\beta' \neq \beta$ with properties (1) and (2).

If $\beta' > \beta$, as β is the least upper bound of S β' cannot satisfy property (2) $\Rightarrow$ ↯

If $\beta' < \beta$, then as β' satisfies (1) we would have that $\beta \neq \sup S$ $\Rightarrow$ ↯

Thus, $\beta = \beta'$ yielding uniqueness. ∎

Some notation

We will often define a set S by writing

$$S = \{x \mid \cdots\},$$

which means that S consists of all x that satisfy the conditions to the right of the vertical bar.

For instance, if we let S denote the set of negative real numbers, we might write

$$S = \{x \in \mathbb{R} \mid x < 0\},$$

or if it is clear from context that S is a subset of $\mathbb{R}$ then we may write $S = \{x \mid x < 0\}$.

and if we let S_1 denote the set of negative integers we could write

$$\begin{aligned} S_1 &= \{x \in \mathbb{R} \mid x < 0 \text{ and } x \text{ is an integer}\} \\ &= \{x \in \mathbb{R} \mid x \text{ is a negative integer}\} \\ &= \{x \in \mathbb{Z} \mid x < 0\} \end{aligned}$$

The Archimedean Property

The property of the real numbers described in the next theorem is called the **Archimedean property**.

Intuitively, it states that it is possible to exceed any positive number, no matter how large, by adding an arbitrary positive number, no matter how small, to itself sufficiently many times.

This is an important property, as it

- ensures every real number can be approximated arbitrarily well by rationals, and
- allows us to rigorously define what we mean by a limit, and hence integration and differentiation.

Theorem (Archimedean Property)
If ρ and ϵ are positive real numbers, then there exists a natural number n such that $n\epsilon > \rho$.

Proof: Suppose that the statement is false.

In which case ρ is an upper bound

for the set

$S = \{x \in \mathbb{R} \mid x = n\epsilon \text{ for some integer } n\}$.

By completeness, S has a supremum β, and

so $n\epsilon \leq \beta$ for all integers n, or equivalently

$(n+1)\epsilon \leq \beta$ for all integers n.

However, $(n+1)\epsilon \leq \beta$ for all integers n

if and only if $n\epsilon \leq \beta - \epsilon$ for all integers n.

i.e. $\beta - \epsilon$ is an upper bound for S.

Really a contradiction to $\beta = \sup S$. ∎

Density of the Rationals and Irrationals

Definition
A set D is **dense in the reals** if every open interval (a, b) contains an element of D.

non-empty

$= \{x \in \mathbb{R} : a < x < b\}$ $\quad a, b \in \mathbb{R}$.

Theorem
The rational numbers are dense in the reals, that is, if a and b are real numbers with $a < b$, then there is a rational number p/q such that $a < p/q < b$.

Proof: Within the Archimedian Property set $p = 1$ and $\varepsilon = b - a$. From this property we know that there exists an integer q such that $q\varepsilon > p$ or equivalently $q(b-a) > 1$.

We also exists an integer j with $j > qa$.

- If $a < 0$ since $q > 0$ we choose $j = 1$.
- If $a > 0$ we set $\varepsilon = 1$ and $p = qa$ in the Archimedean Property.

Let p be the smallest integer such that $p > qa$

In which case $p - 1 \leq qa$.

Exercise: Show that such a p exists.

Hint: Consider the set $\{t \in \mathbb{Z} : t \leq qa\}$

i.e. $\quad qa < p \leq qa + 1$

Since $q(b-a) > 1$

$$qa < p \leq qa + 1 < qa + q(b-a) = qb$$

$$\Rightarrow \quad a < p/q < b.$$

∎

Example
The rational number system is not complete.

Namely, a set of rational numbers may be bounded above by rationals but not have a supremum which is rational.

To see this, consider the set

$$S = \{ r \in \mathbb{Q} \mid r^2 < 2 \}.$$

Thus, if $r \in S$, then $r < \sqrt{2}$.

our previous theorem implies that if $\varepsilon > 0$, then there is a rational number r_0 such that $\sqrt{2} - \varepsilon < r_0 < \sqrt{2}$.

Thus, $\sup S = \sqrt{2}$. However, $\sqrt{2}$ is irrational

Therefore, if r_1 is any rational upper bound of S then $\sqrt{2} < r_1$.

By the denseness of the rationals in $\mathbb{R}$ we know that there exists a rational r_2 with $\sqrt{2} < r_2 < r_1$

Since r_2 would also be a upper bound for S this shows that S has no rational supremum.

Exercise: Prove that $\sqrt{2}$ is irrational.

Prof that $\sqrt{2}$ is not rational.

For Suppose there exist integers p, q with $\gcd(p,q) = 1$ and such that $\sqrt{2} = \frac{p}{q}$

$\Rightarrow \sqrt{2}q = p \Rightarrow 2q^2 = p^2 \Rightarrow 2 \mid p$

$\Rightarrow p = 2k$ for some integer k

$\Rightarrow 2q^2 = 4k^2 \Rightarrow q^2 = 2k^2 \Rightarrow 2 \mid q$

$\Rightarrow \gcd(p,q) \neq 1 \Rightarrow \lightning$. ∎

Theorem
The set of irrational numbers are dense in the reals, that is, if a and b are real numbers with $a < b$, then there is a irrational number t such that $a < t < b$.

Proof: By the denseness of the rationals in $\mathbb{R}$ there exist rationals r_1 and r_2 such that

$$a < r_1 < r_2 < b$$

Let t be the irrational number

$$t = r_1 + \frac{1}{\sqrt{2}}(r_2 - r_1)$$

By construction $r_1 < t < r_2$ and so $a < t < b$

Moreover t is irrational. ∎

Exercise: Prove that t is irrational.

Proof that t is irrational.

For suppose that t is rational

$\Rightarrow$ $\exists$ integers p, q with $\gcd(p, q) = 1$

s.t. $t = \frac{p}{q}$ $\Rightarrow$ $\frac{p}{q} = r_1 + \frac{1}{\sqrt{2}}(r_2 - r_1)$

$$\Rightarrow \frac{p}{q} - r_1 = \frac{1}{\sqrt{2}}(r_2 - r_1)$$

$$\Rightarrow \sqrt{2} = \frac{r_2 - r_1}{\frac{p}{q} - r_1}$$

$\Rightarrow$ $\sqrt{2}$ is rational $\Rightarrow$ ↯ ∎

Infimum of a set

A set S of real numbers is **bounded below** if there is a real number a with $x \geq a$ whenever $x \in S$.

In this case, a is a **lower bound** of S. If a is a lower bound of S, so is any smaller number, because of property of transitivity of our order relation $<$.

If α is a lower bound of S, but no number greater than α is, then α is an **infimum** of S, and we write

$$\alpha = \inf S.$$

i.e. α is the greatest lower bound of S.

Geometrically, this means that there are no points of S to the left of α, but there is at least one point of S to the left of any number greater than α.

Theorem

If a nonempty set S of real numbers is bounded below, then $\inf S$ is the unique real number α such that

(a) $x \geq \alpha$ for all x in S, and

(b) if $\epsilon > 0$, no matter how small, there exists an x_0 in S with $x_0 < \alpha + \epsilon$.

Proof: Exercise ∎

Defn: A set is called bounded if there are numbers a and b such that $a \leq x \leq b$ for all $x \in S$.

Thus a bounded set has a unique infimum and a unique supremum.

Exercise: Prove that if S is a bounded non-empty set, then $\inf S \leq \sup S$. Moreover, give an example of a bounded set with $\inf S = \sup S$ and an example of a set S with $\inf S \neq \sup S$.

The extended real number system

A nonempty set S of real numbers is **unbounded above** if it has no upper bound, and **unbounded below** if it has no lower bound.

It is convenient to adjoin to the real number system two fictitious points, $+\infty$, which we usually write more simply as ∞, and $-\infty$, and to define the order relationships between them and any real number x by

$$-\infty < x < \infty.$$

We call ∞ and $-\infty$ **points at infinity**.

If S is a nonempty set of reals, we write

$$\sup S = \infty$$

to indicate that S is unbounded above, and

$$\inf S = -\infty$$

to indicate that S is unbounded below.

Example

- If $S = \{x \in \mathbb{R} \mid x < 2\}$, then $\sup S = 2$ and $\inf S = -\infty$ Note $\sup S \notin S$
- If $S = \{x \in \mathbb{R} \mid x \geq -2\}$, then $\sup S = \infty$ and $\inf S = -2$ Note $\inf S \in S$.
- If $S = \emptyset$ then $\sup S = -\infty$ and $\inf S = \infty$.

Every real number is an upper bound for $\emptyset$.

Every real number is a lower bound for $\emptyset$.

The real number system with ∞ and $-\infty$ adjoined is called the **extended real number system**, or simply the *extended reals*.

A member of the extended reals differing from $-\infty$ and ∞ is called **finite**, that is, an ordinary real number is finite.

The arithmetic relationships among ∞, $-\infty$, and the real numbers are defined as follows.

(a) If a is any real number, then

$$\begin{aligned} a+\infty &= \infty + a = \infty, \\ a-\infty &= -\infty + a = -\infty, \\ \frac{a}{\infty} &= \frac{a}{-\infty} = 0. \end{aligned}$$

(b) If $a > 0$, then

$$a \cdot \infty = \infty \cdot a = \infty, \quad \text{and} \quad a \cdot (-\infty) = (-\infty) \cdot a = -\infty.$$

(c) If $a < 0$, then

$$a \cdot \infty = \infty \cdot a = -\infty \quad \text{and} \quad a \cdot (-\infty) = (-\infty) \cdot a = \infty.$$

We also define

$$\infty + \infty = \infty \cdot \infty = (-\infty) \cdot (-\infty) = \infty$$

and

$$-\infty - \infty = \infty \cdot (-\infty) = (-\infty) \cdot \infty = -\infty.$$

Finally, we set

$$|\infty| = |-\infty| = \infty.$$

It is not useful to define $\infty - \infty$, $0 \cdot \infty$, ∞/∞, and $0/0$. These are sometimes called **indeterminate forms**, and left undefined.

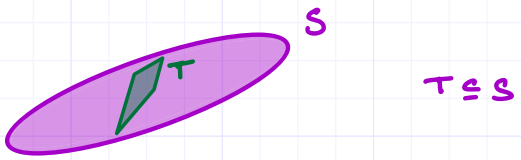
Note: The introduction of ∞ and $-\infty$ along with the arithmetic and order relations defined above leads to simplifications in the statements of theorems we will encounter.

Some set theory and the topology of $\mathbb{R}$

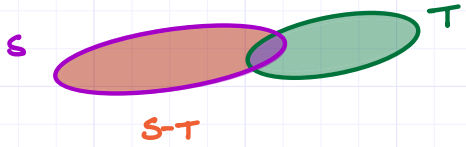
The following definition applies to arbitrary sets, with the understanding that the members of all sets under consideration in any given context come from a specific collection of elements, called the **universal set**. For most of this course, the universal set is the real numbers.

Definition Let S and T be sets.

(a) We say that S **contains** T, and write $S \supseteq T$ or $T \subseteq S$, if every member of T is also in S. In this case, T is called a **subset** of S.



(b) We denote by $S - T$ the set of elements that are in S but not in T.

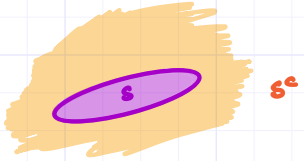


(c) We say that S **equals** T, and we write $S = T$, if S contains T and T contains S. Thus, $S = T$ if and only if S and T have the same members.

(d) We say that S **strictly contains** T, if S contains T but T does not contain S, that is, if every member of T is also in S, but at least one member of S is not in T.

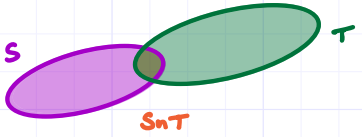
Here we might sometimes write $T \subset S$

(e) The **complement** of S, denoted by S^c, is the set of elements in the universal set that are not in S.



(f) The **union** of S and T, denoted by $S \cup T$, is the set of elements in at least one of S and T.

(g) The **intersection** of S and T, denoted by $S \cap T$, is the set of elements in both S and T. If $S \cap T = \emptyset$, the empty set, then S and T are said to be **disjoint sets**.



(h) A set with only one member is called a **singleton set**.

or singleton for short.

Example

Let $S = \{x \in \mathbb{R} \mid 0 < x < 1\}$, $T = \{x \in \mathbb{R} \mid 0 < x < 1 \text{ and } x \text{ is rational}\}$.

and $U = \{x \in \mathbb{R} \mid 0 < x < 1 \text{ and } x \text{ is irrational}\}$.

Then $U \subset S$, $T \subset S$, $T \cup U = S$, $T \cap U = \emptyset$, $U \cap S = U$ and $T \cap S = T$

Also $S - U = T$ and $S - T = U$.

Every set S contains the empty set $\emptyset$. For if $\emptyset$ is was not contained in S, this means that some member of $\emptyset$ is not in S. However, as $\emptyset$ has no members, this is a cannot happen.

If S is any set, then

$$(S^c)^c = S \quad \text{and} \quad S \cap S^c = \emptyset.$$

If S is a set of real numbers, then $S \cup S^c = \mathbb{R}$.

The definitions of union and intersection have generalisations: If $\mathcal{F}$ is an arbitrary collection of sets, then $\cup\{S \mid S \in \mathcal{F}\}$ is the set of all elements that are members of at least one of the sets in $\mathcal{F}$.

Similarly, $\cap\{S \mid S \in \mathcal{F}\}$ is the set of all elements that are members of every set in $\mathcal{F}$.

The union and intersection of finitely many sets $S_1, \ldots, S_n$ are also written as

$$\bigcup_{k=1}^{n} S_k \quad \text{and} \quad \bigcap_{k=1}^{n} S_k.$$

The union and intersection of an infinite sequence $\{S_k\}_{k=1}^{\infty}$ of sets are written as

$$\bigcup_{k=1}^{\infty} S_k \quad \text{and} \quad \bigcap_{k=1}^{\infty} S_k.$$

Here we are taking our universal set to be $\mathbb{R}$.

This is sometimes written as $\bigcup_{S \in \mathcal{F}} S$

This is sometimes written as $\bigcap_{S \in \mathcal{F}} S$

we might also adopt the notation $\bigcup_{k \in \{1,\ldots,n\}} S_k$

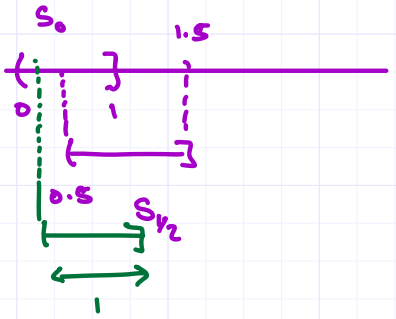
Example

- Let $\mathcal{F}$ be the collection of sets

$$S_p = \{ x \in \mathbb{R} : p < x \leq 1+p \} \qquad 0 < p \leq 1/2$$

i.e.

$$\mathcal{F} = \{ S_p : 0 < p \leq 1/2 \}$$



Then $\bigcup_{S \in \mathcal{F}} S = \{ x \in \mathbb{R} : 0 < x \leq 3/2 \}$

$\bigcap_{S \in \mathcal{F}} S = \{ x \in \mathbb{R} : 0.5 < x \leq 1 \}$

- If for each positive integer k, we let S_k to be the set of real numbers which can be written as m/k for some integer m, i.e. $S_k = \{ m/k \mid m \in \mathbb{Z} \}$, then

$$\bigcup_{k=1}^{\infty} S_k = \mathbb{R}.$$

If a and b are in the extended reals and $a < b$, then the **open interval** (a, b) is defined by

$$(a, b) = \{x \in \mathbb{R} \mid a < x < b\}.$$

The open intervals (a, ∞) and $(-\infty, b)$ are **semi-infinite** if a and b are finite, and $(-\infty, \infty)$ is the entire real line $\mathbb{R}$.

Definition

(a) If x_0 is a real number and $\epsilon > 0$, then the open interval $(x_0 - \epsilon, x_0 + \epsilon)$ is called an **ϵ-neighborhood** of x_0.

(b) If a set S contains an ϵ-neighborhood of x_0, then S is called a **neighborhood** of x_0, and x_0 is said to be an **interior point** of S.

(c) The set of interior points of S is the **interior** of S, denoted by S^0.

(d) If every point of S is an interior point (that is, $S^0 = S$), then S is said to be **open**.

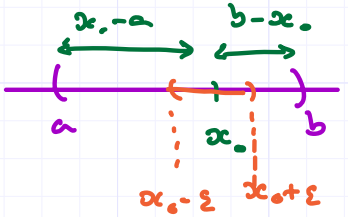
(e) A set S is **closed** if S^c is open.

The idea of a neighborhood is fundamental and occurs in many other contexts. Whatever the context, the idea is the same: some definition of **closeness** is given (for example, two real numbers are **close** if their difference is **small**), and a neighborhood of a point x_0 is a set that contains all points sufficiently close to x_0.

Example

- An open interval (a,b) is an open set, because if $x_0 \in (a,b)$ and $\varepsilon \leq \min\{x_0 - a, b - x_0\}$ then

$$(x_0 - \varepsilon, x_0 + \varepsilon) \subseteq (a,b)$$

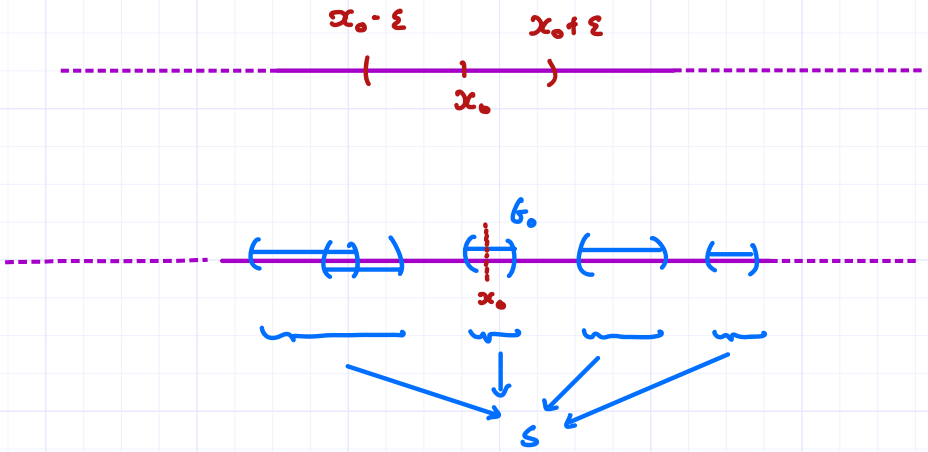


- The entire real line $\mathbb{R}$ is open, since if $x_0 \in \mathbb{R}$ then $x_0 \in (x_0 - 1, x_0 + 1) \subseteq \mathbb{R}$.

- Therefore, $\emptyset = \mathbb{R}^c$ is closed.

- We also observe that $\emptyset$ is open, since the empty set contains no elements it vacuously contains all neighbourhoods around all its points.
- Thus, $\mathbb{R} = \emptyset^c$ is closed.
- Note $\mathbb{R}$ and $\emptyset$ are the only subsets of $\mathbb{R}$ which are both open and closed.

An example of a deleted neighborhood is $(x_0 - \varepsilon, x_0) \cup (x_0, x_0 + \varepsilon)$



Definition

A **deleted neighborhood** of a point x_0 is a set that contains every point of some neighborhood of x_0 except for x_0 itself.

Theorem

(a) An arbitrary union of open sets is open.

(b) An arbitrary intersection of closed sets is closed.

Proof: (a) Let $\mathcal{G}$ be a collection of open sets and let

$$S = \bigcup_{G \in \mathcal{G}} G$$

If $x_0 \in S$, then $x_0 \in G_0$ for some $G_0 \in \mathcal{G}$.

Since G_0 is open, it contains some ε-neighbourhood of x_0.

Moreover, since $G_0 \subseteq S$, this ε-neighbourhood of x_0 is contained in S, and hence S is open.

(b) Let $\mathcal{F}$ be a collection of closed sets and let $T = \bigcap_{F \in \mathcal{F}} F$ by De Morgan's Formula

Exercise Sheet 1 : Question 6

$$T^c = \bigcup_{F \in \mathcal{F}} F^c$$

and since F^c is open, Part (a) implies that T^c is open $\Rightarrow T = (T^c)^c$ is closed. ∎

Example

- $a, b \in \mathbb{R}$ with $a < b$, the set

$$[a,b] = \{ x \mid a \leq x \leq b \}$$

is closed, since

$$[a,b]^c = (-\infty, a) \cup (b, \infty)$$ is open.

open open

union of open sets

We call $[a,b]$ a closed interval.

- The sets $[a,b)$ and $(a,b]$ are neither open nor closed and are often called half open / closed intervals.

To see that $[a,b)$ is not open, observe that for all $\varepsilon > 0$ the ε-neighbourhood $(a-\varepsilon, a+\varepsilon)$ is not contained in $[a,b)$ and thus $[a,b)$ is not open. We also claim that $[a,b)$ is not closed for suppose it was closed, this would imply that $[a,b)^c = (-\infty, a) \cup [b, \infty)$ is open but by a similar argument we can show that $[a,b)^c$ is not open $\Rightarrow$ ↯

- Semi-infinite closed intervals are sets of the form

$$[a, \infty) = \{ x \in \mathbb{R} \mid a \leq x \}$$

$$(-\infty, a] = \{ x \in \mathbb{R} \mid x \leq a \}$$

These are closed sets, since their complements are open.

Remark: It can be shown that every non-empty open subset of $\mathbb{R}$ is the union of open intervals.

Exercise: Show that the intersection of finitely many open sets is open, and that the union of finitely many closed sets is closed.

Moreover, show that this result does not hold if we exchange finite for infinite.

Definition Let S be a subset of $\mathbb{R}$.

(a) A point $x_0 \in \mathbb{R}$ is a **limit point** of S if every deleted neighborhood of x_0 contains a point of S.

(b) A point $x_0 \in \mathbb{R}$ is a **boundary point** of S if every neighborhood of x_0 contains at least one point in S and one not in S. The set of boundary points of S is called the **boundary** of S and is denoted by ∂S, and the **closure** of S, denoted by $\overline{S}$, is defined by $\overline{S} = S \cup \partial S$.

(c) A point $x_0 \in \mathbb{R}$ is an **isolated point** of S if $x_0 \in S$ and there is a neighborhood of x_0 that contains no other point of S.

(d) A point $x_0 \in \mathbb{R}$ is **exterior** to S if x_0 is in the interior of S^c. The collection of such points is the **exterior** of S.

Can replace neighbourhood by ε-neighbourhood.

Example $S = (-\infty, -1] \cup (1,2) \cup \{3\}$

(a) The set of limit points of S is $(-\infty, -1] \cup [1,2]$.

3 is not a limit point since $(2.5, 3.5) \cap S = \{3\}$

if $x \in (-\infty, -1]$, then for all $\varepsilon > 0$, $(x-\varepsilon, x+\varepsilon) \cap S \supseteq (x-\varepsilon, x)$

if $x \in [1, 3/2]$, then for all $\varepsilon > 0$, either

$x + \varepsilon/2 \in (x-\varepsilon, x+\varepsilon) \cap S$ or $3/2 \in (x-\varepsilon, x+\varepsilon) \cap S$.

Similarly, if $x \in [3/2, 2]$, then $\forall \varepsilon > 0$ either

$x - \varepsilon/2 \in (x-\varepsilon, x+\varepsilon) \cap S$ or $3/2 \in (x-\varepsilon, x+\varepsilon) \cap S$.

(b) $\partial S = \{-1, 1, 2, 3\}$

$\Rightarrow \overline{S} = S \cup \partial S = (-\infty, -1] \cup [1,2] \cup \{3\}$.

observe that no interior point of S can be a boundary point

Moreover, $\{-1, 1, 2, 3\} \subseteq \partial S$

e.g. $-1 \in S \cap (-1-\varepsilon, -1+\varepsilon)$

and $-1 + \varepsilon/2 \in S^c$ $\forall \varepsilon \in (0,1)$

Exercise: prove that $\{1, 2, 3\} \subseteq \partial S$.

To show that $\partial S = \{-1, 1, 2, 3\}$

we observe that

$((-\infty, -1] \cup [1,2] \cup \{3\})^c$

$= (-1, 1) \cup (2, 3) \cup (3, \infty)$

which is an open set and so no element of this set is a boundary point of S.

Recall: $S = (-\infty, -1] \cup (1,2) \cup \{3\}$.

(c) The only isolated point of S is $\{3\}$.

(c) A point $x_0 \in \mathbb{R}$ is an **isolated point** of S if $x_0 \in S$ and there is a neighborhood of x_0 that contains no other point of S.

using our previous arguments if

$x \in (-\infty, -1] \cup [1,2)$

or if $x \in (-1,1) \cup (2,3) \cup (3, -\infty)$

we see that x is not an isolated point

Thus the only case to check is $x = 3$.

However, $(3 - \frac{1}{2}, 3 + \frac{1}{2}) \cap S = 3$

$\Rightarrow$ 3 is an isolated point of S.

(d) $(S^c)^\circ = (-1,1) \cup (2,3) \cup (3, \infty)$

(d) A point $x_0 \in \mathbb{R}$ is **exterior** to S if x_0 is in the interior of S^c. The collection of such points is the **exterior** of S.

observe that $S^c = (-1,1] \cup [2,3) \cup (3, \infty)$

we know $(-1,1]^\circ = (-1,1)$

$[2,3)^\circ = (2,3)$

and $(3, \infty)$ is open

$\Rightarrow (S^c)^\circ = (-1,1) \cup (2,3) \cup (3, \infty)$

Exercise: Find the closure of $\mathbb{Q}$ and the boundary of $\mathbb{Q}$.

A point $x_0 \in \mathbb{R}$ is a **limit point** of S if every deleted neighborhood of x_0 contains a point of S.

Theorem

A set S is closed if and only if no point of S^c is a limit point of S.

Corollary

A set is closed if and only if it contains all its limit points.

} if no point of S^c is a limit point of S then S needs to contain all of its limit points.

Proof of Thm:

"$\Rightarrow$" Suppose that S is closed, and $x_0 \in S^c$.
Since S is closed, this implies that S^c is open
thus there exists a neighborhood of x_0
contained in S^c and hence x_0 is not a limit
point of S.

"$\Leftarrow$" Now suppose that no point of S^c is a limit
point of S. In which case every point in S^c
has a neighborhood contained in S^c.
Thus, S^c is open $\Rightarrow$ S is closed ∎

Open coverings

A collection $\mathcal{H}$ of open sets is an **open covering** of a set S if every point in S is contained in a set H belonging to $\mathcal{H}$, that is,

$$S \subset \cup\{H \mid H \in \mathcal{H}\}. \quad = \quad \bigcup_{H \in \mathcal{H}} H$$

Example

- $S = [0,1]$.

Let $n \in \mathbb{N}$ and consider

$$\mathcal{H}_n = \left\{ \left(x - \tfrac{1}{n}, x + \tfrac{1}{n}\right) \mid x \in (0,1) \right\}$$ ← uncountable

} elements of $\mathcal{H}$ are open intervals.

$\mathcal{H}_n$ is an open cover of S.

Indeed, every elmt of $\mathcal{H}_n$ is an open intrval and

$$\bigcup_{H \in \mathcal{H}_n} H = \bigcup_{x \in (0,1)} \left(x - \tfrac{1}{n}, x + \tfrac{1}{n}\right) = \left(-\tfrac{1}{n}, 1 + \tfrac{1}{n}\right) \supseteq [0,1]$$

And open cover would be ← Countable

$$G_n = \left\{ \left(q - \tfrac{1}{n}, q + \tfrac{1}{n}\right) \mid q \in \mathbb{Q} \right\}$$

or

$$\left\{ \left(-\tfrac{1}{2}, \tfrac{1}{2}\right), (0,1), \left(\tfrac{1}{2}, \tfrac{3}{2}\right) \right\}.$$ ← finite.

Subsets of $\mathbb{R}$ which are closed and bounded are sometimes called compact

} The converse also holds true

See Exercise Sheet 1: Question 11.

This is a general notion from topology.

Theorem (Heine-Borel Theorem)

If $\mathcal{H}$ is an open covering of a closed and bounded subset S of the real line, then S has an open covering $\widetilde{\mathcal{H}}$ consisting of finitely many open sets belonging to $\mathcal{H}$.

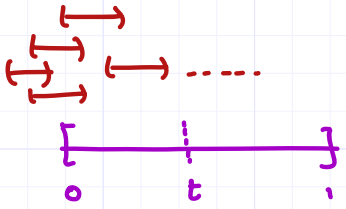
Proof: We will prove the statement for $[0,1]$

Exercise: How can one extend the proof of $[0,1]$ to an arbitrary closed interval $[a,b]$

Exercise: How can one extend the solution to the above exercise to an arbitrary closed ball.

Proof that any open cover $\mathcal{H}$ of $[0,1]$ can be reduced to a finite open cover of $[0,1]$.

Let $F = \{ t \in [0,1] \mid [0,t]$ can be covered by finitely many sets from $\mathcal{H} \}$.



$t \in F$ if we can find $U_1, U_2, \dots U_n \in \mathcal{H}$ such that $\bigcup_{i=1}^{n} U_i \supseteq [0,t]$.

The result will follow if we can show that $1 \in F$.

Observe that $0 \in F$ since $[0,0] = \{0\}$

and if $[0,1] \subseteq \bigcup_{H \in \mathcal{H}} H \Rightarrow \exists H \in \mathcal{H}$ s.t. $0 \in H$.

i.e. we can find a single element of $\mathcal{H}$ which contains 0.

Moreover, by construction for all $t \in F$ we have $0 \leq t \leq 1$.

Thus F is non-empty and is bounded above and below.

By the completeness axiom, we have that F has a Supremum.

Let us denote the Supremum of F by γ, i.e. $\gamma = \sup F$.

Exercise: Prove that $\gamma \in F$.

Now suppose that $\gamma < 1$

then there exists $U_1, U_2 \dots U_n$

such that

$$\bigcup_{i=1}^{n} U_i \supseteq [0, \gamma].$$

$\Rightarrow$ there exists a $j \in \{1, \dots, n\}$

such that $\gamma \in U_j$

However, U_j is open and so $\exists\, \varepsilon > 0$

such that $(\gamma - \varepsilon, \gamma + \varepsilon) \subseteq U_j$

$\Rightarrow$

$$\bigcup_{i=1}^{n} U_i \supseteq [0, \gamma + \varepsilon/2]$$

$\Rightarrow$ $\{U_1 \dots U_n\}$ is a finite collection of sets from $\mathcal{H}$

which cover $[0, \gamma + \varepsilon/2]$, i.e. $\gamma + \varepsilon/2 \in F$

But $\gamma + \varepsilon/2 > \gamma$ $\Rightarrow$ ↯

Suppose $\gamma \notin F$. Since $\mathcal{H}$ is an open cover of $[0,1]$.

$\Rightarrow \exists\, H \in \mathcal{H}$ s.t. $\gamma \in H$. Since H is open

$\Rightarrow \exists\, \varepsilon > 0$ s.t. $(\gamma - \varepsilon, \gamma + \varepsilon) \subseteq H$

By defn of $\sup F$ $\Rightarrow \exists\, t_0 \in F$ s.t. $\gamma - \varepsilon < t_0$

By defn of F $\Rightarrow \exists\, U_1, U_2 \dots U_n \in \mathcal{H}$ s.t.

$$\bigcup_{i=1}^{n} U_i \supseteq [0, t_0] \Rightarrow \{U_1, U_2 \dots U_n, H\} \text{ is}$$

a finite collection of open sets from $\mathcal{H}$ which cover $[0, \gamma]$.

∎

A point $x_0 \in \mathbb{R}$ is a **limit point** of S if every deleted neighborhood of x_0 contains a point of S.

Theorem (Bolzano-Weierstrass Theorem)

Every bounded infinite set of real numbers has at least one limit point.

} A very important result and a consequence of the Heine-Borel Theorem.

Proof: To prove this result we will show that a bounded set without a limit point contains a finite number of points.

To this end let S be a bounded set without a limit point.

If S has no limit points, then it trivially contains all of its limit points. Hence, S is closed.

Since S has no limit points, to every $x \in S$ there exists an $\varepsilon_x > 0$ such that $(x - \varepsilon_x, x + \varepsilon_x) \cap S = \{x\}$.

Hence $\{(x - \varepsilon_x, x + \varepsilon_x) \mid x \in S\}$ is an open cover of S.

Now S is closed and bounded and so its open cover reduces to finite open cover i.e. $\exists\ x_1, x_2 \dots x_m$ such that

$$\bigcup_{i=1}^{m} (x_i - \varepsilon_{x_i}, x_i + \varepsilon_{x_i}) \supseteq S.$$

but each of these sets contains a single element of S

Thus S is finite. ∎

2 – Differential Calculus of Functions

In this chapter we study the differential calculus of functions of one variable.

- § 2.1 introduces the concept of function and discusses arithmetic operations on functions, limits, one-sided limits, limits at $\pm\infty$, and monotonic functions.
- § 2.2 defines continuity and discusses removable discontinuities, composite functions, bounded functions, the intermediate value theorem, uniform continuity, and additional properties of monotonic functions.
- § 2.3 introduces the derivative and its geometric interpretation. Topics covered include the interchange of differentiation and arithmetic operations, the chain rule, one-sided derivatives, extreme values of a differentiable function, Rolle's theorem, the intermediate value theorem for derivatives, and the mean value theorem and its consequences.

2.1 Functions and Limits

Functions

- A rule f that assigns to each member of a nonempty set D a unique member of a set Y is a **function from D to Y**.
- We write the relationship between a member x of D and the member y of Y that f assigns to x as

$$y = f(x).$$

- The set D is called the **domain** of f, denoted by D_f.
- The members of Y are the possible **values** of f.
- If $y_0 \in Y$ and there is an x_0 in D such that $f(x_0) = y_0$ then we say that f **attains**, or **assumes**, the value y_0.
- The set of values attained by f is the **range** of f.
- A **real-valued function of a real variable** is a function whose domain and range are both subsets of the reals.

Y is sometimes referred to as the co-domain of f.

Note the range and codomain do not need to coincide.

Example

- $f : \mathbb{R} \to [0,\infty)$ $\quad f(x) = x^2$ $\quad$ Range$(f) = [0,\infty)$

  Domain codomain rule

  Note: Strictly speaking the function $g : \mathbb{R} \to \mathbb{R}$ $\quad g(x) = x^2$ is classed as a different function to f.

  Domain codomain rule

  Same can be function $h : [0,1] \to \mathbb{R}$ $\quad h(x) = x^2$.

- Note the rule "$(f(x))^2 = x$" does not define a function except on the subset $\{0\}$.

  However, the rule "$(f(x))^2 = x$ with $f(x) > 0$" does define a function with $D_f = [0,\infty)$ and codomain $[0,\infty)$.

Definition

If $D_f \cap D_g \neq \emptyset$, then $f+g$, $f-g$, and $f \cdot g$ are defined on $D_f \cap D_g$ by

$$(f+g)(x) = f(x) + g(x),$$
$$(f-g)(x) = f(x) - g(x),$$
$$(f \cdot g)(x) = f(x) \cdot g(x).$$

The quotient f/g also is defined by

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$$

for x in $D_f \cap D_g$ such that $g(x) \neq 0$.

Example

- Domain, codomain, Domain, codomain

  $f : [-2,2] \to \mathbb{R}$ $\quad$ $g : [1,\infty) \to \mathbb{R}$

  $f(x) = \sqrt{4-x^2}$ $\quad$ $g(x) = \sqrt{x-1}$

  rules.

  Range of f: $[0,2]$

  Range g: $[0,\infty)$

  $f+g$, $f-g$, $f \cdot g$ are all defined on $[1,2]$ i.e. Domain of $f+g$ is $[1,2]$.

  More explicitly $f+g : [1,2] \to \mathbb{R}$

  $(f+g)(x) = f(x) + g(x) = \sqrt{4-x^2} + \sqrt{x-1}$

  $\frac{f}{g}$ is defined on $(1,2]$ i.e. Domain of $\frac{f}{g}$ is $(1,2]$

  More explicitly $\frac{f}{g} : (1,2] \to \mathbb{R}$

  $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \frac{\sqrt{4-x^2}}{\sqrt{x-1}}$

Limits

Definition

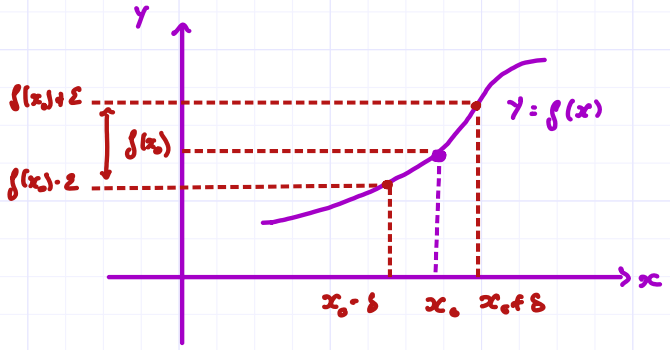
We say that $f(x)$ **approaches the limit** L **as** x **approaches** x_0, and write

$$\lim_{x \to x_0} f(x) = L,$$

if f is defined on some deleted neighborhood of x_0 and, for every $\epsilon > 0$, there is a $\delta > 0$ such that if $|x - x_0| < \delta$, then

$$|f(x) - L| < \epsilon.$$

$x \in D_f$ - Domain of f.



Example

- Let $c \in \mathbb{R}$ with $c \neq 0$ and define $f : \mathbb{R} \to \mathbb{R}$ by $f(x) = c \cdot x$.

Claim: For $x_0 \in \mathbb{R}$

$$\lim_{x \to x_0} f(x) = f(x_0) = cx_0.$$

↓ follows by definition.

Note that $L = cx_0$

Note: It is not always the case that $\lim_{x \to x_0} f(x) = f(x_0)$

Proof of claim: Let $\varepsilon > 0$ and set $\delta = \varepsilon / |c|$

To be determined at the end of our calculation.

If $x \in \mathbb{R}$ is such that $|x - x_0| < \delta$, then

$$|f(x) - cx_0| = |cx - cx_0|$$
$$= |c|\,|x - x_0|$$
$$< |c|\delta \leq |c|\,\varepsilon/|c| = \varepsilon. \quad \blacksquare$$

- Defn $f : \mathbb{R} - \{0\} \to \mathbb{R}$ by $f(x) = x \sin(1/x)$

Claim: $\lim_{x \to 0} f(x) = 0$ — observe that $f(0)$ is not defined

↓ since $0 \notin \text{Domain}(f)$

Here $L = 0$ and $x_0 = 0$

Proof of claim: Let $\varepsilon > 0$ be given and let $\delta = \varepsilon$ — To be determined at the end of our calculation.

If $x \in \mathbb{R} - \{0\}$ is such that $\underbrace{|x - 0|}_{|x_0|} < \delta$, then

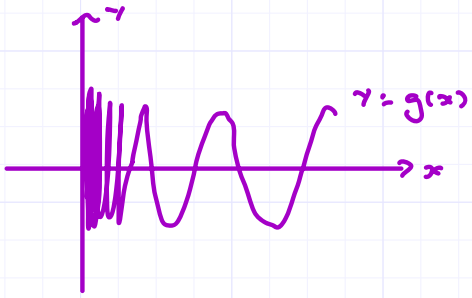
$$|f(x) - 0| = |f(x)|$$

$$= |x \sin(x)| = |x|\,|\sin(1/x)|$$

$$\leq |x| \cdot 1 = |x| < \varepsilon \quad \blacksquare$$

- Defn $g : \mathbb{R} - \{0\} \to \mathbb{R}$ defined by $g(x) = \sin(1/x)$

Claim: The limit $\lim_{x \to 0} g(x)$ does not exist.



Proof of claim: Observe that, for $n \in \mathbb{N}$,

$$g(1/2\pi n) = \sin(1/1/2\pi n) = \sin(2\pi n) = 0$$

and

$$g(2/4\pi n + \pi) = \sin\left(1/2/4\pi n + \pi\right) = \sin\left(\frac{4\pi n + \pi}{2}\right) = \sin\left(2\pi n + \frac{\pi}{2}\right) = \sin\left(\frac{\pi}{2}\right) = 1$$

i.e. we can find points arbitrarily close to 0 which get mapped to 0 by g but also points arbitrarily close to 0 which get mapped to 1 by g.